Input system:
Dependent variables:x_1, x_2, x_3
Independent variables:t

$$\frac{dx_1}{dt} = x_3^3\sin(\cos(a\ln(x_2)^2+bx_3))+x_3^3b\ln(x_2)^4+\sin(a\ln(x_2)^2+bx_3)^5+\mathrm{Dv}(g^2,1,(-1);x_1) \quad (1)$$

$$\frac{dx_2}{dt} = x_2^2\cos(\sin(a\ln(x_2)^2+bx_3))+\cos(a\ln(x_2)^2+bx_3)^4+\mathrm{EK}(\sin(\sin(a\ln(x_2)^2+bx_3)),x_1) \quad (2)$$

$$\frac{dx_3}{dt} = x_1\mathrm{ch}(a\ln(x_2)^2+bx_3)^2+x_1\sin(a\ln(x_2)^2+bx_3)+\mathrm{sh}(a\ln(x_2)^2+bx_3)^5 \quad (3)$$

==========

Result:
Dependent variables:$s_4, s_{15}, x_3, s_{11}, s_{14}, x_2, s_2, s_{16}, s_{12}, s_{13}, s_8, s_3, s_6, x_1, s_1, s_{10}, s_5, s_7, s_9$
Independent variables:t
Additional variables:

$$s_1 = \ln(x_2) \quad (4)$$

$$s_2 = \mathrm{inv}(x_2) \quad (5)$$

$$s_3 = \cos(as_1^2+bx_3) \quad (6)$$

$$s_4 = \sin(as_1^2+bx_3) \quad (7)$$

$$s_5 = \sin(s_3) \quad (8)$$

$$s_6 = \cos(s_3) \quad (9)$$

$$s_7 = \mathrm{Dv}(g^2,1,(-1);x_1) \quad (10)$$

$$s_8 = \mathrm{DV}(g^2,1,(-1);x_1) \quad (11)$$

$$s_9 = \cos(s_4) \quad (12)$$

$$s_{10} = \sin(s_4) \quad (13)$$

$$s_{11} = \mathrm{EK}(s_{10},x_1) \quad (14)$$

$$s_{12} = \mathrm{EKs}(s_{10},x_1) \quad (15)$$

$$s_{13} = \mathrm{EKc}(s_{10},x_1) \quad (16)$$

$$s_{14} = \mathrm{EKi}(s_{10},x_1) \quad (17)$$

$$s_{15} = \mathrm{ch}(as_1^2+bx_3) \quad (18)$$

$$s_{16} = \mathrm{sh}(as_1^2+bx_3) \quad (19)$$

$$\frac{dx_1}{dt} = x_3^3bs_1^4+s_4^5+x_3^3s_5+s_7 \quad (20)$$

$$\frac{dx_2}{dt} = s_3^4+x_2^2s_9+s_{11} \quad (21)$$

$$\frac{dx_3}{dt} = s_{16}^5+x_1s_{15}^2+x_1s_4 \quad (22)$$

$$\frac{ds_1}{dt} = s_2s_3^4+s_2x_2^2s_9+s_2s_{11} \quad (23)$$

$$\frac{ds_2}{dt} = (-1)s_2^2 s_3^4 + (-1)s_2^2 x_2^2 s_9 + (-1)s_2^2 s_{11} \tag{24}$$

$$\frac{ds_3}{dt} = (-2)s_4 a s_1 s_2 s_3^4 + (-2)s_4 a s_1 s_2 x_2^2 s_9 + (-1)s_4 b s_{16}^5 + (-2)s_4 a s_1 s_2 s_{11} + (-1)s_4 b x_1 s_{15}^2 + (-1)s_4^2 b x_1 \tag{25}$$

$$\frac{ds_4}{dt} = 2s_3^5 a s_1 s_2 + 2s_3 a s_1 s_2 x_2^2 s_9 + s_3 b s_{16}^5 + 2s_3 a s_1 s_2 s_{11} + s_3 b x_1 s_{15}^2 + s_3 b x_1 s_4 \tag{26}$$

$$\frac{ds_5}{dt} = (-2)s_6 s_4 a s_1 s_2 s_3^4 + (-2)s_6 s_4 a s_1 s_2 x_2^2 s_9 + (-1)s_6 s_4 b s_{16}^5 + (-2)s_6 s_4 a s_1 s_2 s_{11} + (-1)s_6 s_4 b x_1 s_{15}^2 + (-1)s_6 s_4^2 b x_1 \tag{27}$$

$$\frac{ds_6}{dt} = 2s_5 s_4 a s_1 s_2 s_3^4 + 2s_5 s_4 a s_1 s_2 x_2^2 s_9 + s_5 s_4 b s_{16}^5 + 2s_5 s_4 a s_1 s_2 s_{11} + s_5 s_4 b x_1 s_{15}^2 + s_5 s_4^2 b x_1 \tag{28}$$

$$\frac{ds_7}{dt} = s_8 x_3^3 b s_1^4 + s_8 s_4^5 + s_8 x_3^3 s_5 + s_8 s_7 \tag{29}$$

$$\frac{ds_8}{dt} = (-1)x_3^3 b s_1^4 s_7 p_1 x_1^2 + (-1)x_3^3 b s_1^4 s_7 p_2 x_1 + (-1)x_3^3 b s_1^4 s_7 p_3 + (-1)s_4^5 s_7 p_1 x_1^2 + (-1)x_3^3 s_5 s_7 p_1 x_1^2 + (-1)s_4^5 s_7 p_2 x_1 + \tag{30}$$

$$\frac{ds_9}{dt} = (-2)s_{10} s_3^5 a s_1 s_2 + (-2)s_{10} s_3 a s_1 s_2 x_2^2 s_9 + (-1)s_{10} s_3 b s_{16}^5 + (-2)s_{10} s_3 a s_1 s_2 s_{11} + (-1)s_{10} s_3 b x_1 s_{15}^2 + (-1)s_{10} s_3 \tag{31}$$

$$\frac{ds_{10}}{dt} = 2s_9 s_3^5 a s_1 s_2 + 2s_9^2 s_3 a s_1 s_2 x_2^2 + s_9 s_3 b s_{16}^5 + 2s_9 s_3 a s_1 s_2 s_{11} + s_9 s_3 b x_1 s_{15}^2 + s_9 s_3 b x_1 s_4 \tag{32}$$

$$\frac{ds_{11}}{dt} = 2s_{14} s_9 s_3^5 a s_1 s_2 + s_{12} s_{14} x_3^3 b s_1^4 + 2s_{14} s_9^2 s_3 a s_1 s_2 x_2^2 + s_{14} s_9 s_3 b s_{16}^5 + 2s_{14} s_9 s_3 a s_1 s_2 s_{11} + s_{14} s_9 s_3 b x_1 s_{15}^2 + s_{12} s_{14} s \tag{33}$$

$$\frac{ds_{12}}{dt} = 2s_{13} s_{14} s_9 s_3^5 a s_1 s_2 + s_{12} s_{13} s_{14} x_3^3 b s_1^4 + 2s_{13} s_{14} s_9^2 s_3 a s_1 s_2 x_2^2 + s_{13} s_{14} s_9 s_3 b s_{16}^5 + 2s_{13} s_{14} s_9 s_3 a s_1 s_2 s_{11} + s_{13} s_{14} s \tag{34}$$

$$\frac{ds_{13}}{dt} = (-2)s_{12} s_{14} s_9 s_3^5 a s_1 s_2 + (-1)s_{12}^2 s_{14} x_3^3 b s_1^4 + (-2)s_{12} s_{14} s_9^2 s_3 a s_1 s_2 x_2^2 + (-1)s_{12} s_{14} s_9 s_3 b s_{16}^5 + (-2)s_{12} s_{14} s_9 s_3 \tag{35}$$

$$\frac{ds_{14}}{dt} = (-2)s_{10} s_{14}^3 s_{12} s_9 s_3^5 a s_1 s_2 + (-1)x_3^3 b s_1^4 s_{10} s_{12}^2 s_{14}^3 + (-2)s_{10} s_{14}^3 s_{12} s_9^2 s_3 a s_1 s_2 x_2^2 + (-1)s_{10} s_{14}^3 s_{12} s_9 s_3 b s_{16}^5 + (- \tag{36}$$

$$\frac{ds_{15}}{dt} = 2s_{16} a s_1 s_2 s_3^4 + 2s_{16} a s_1 s_2 x_2^2 s_9 + s_{16}^6 b + 2s_{16} a s_1 s_2 s_{11} + s_{16} b x_1 s_{15}^2 + s_{16} b x_1 s_4 \tag{37}$$

$$\frac{ds_{16}}{dt} = 2s_{15} a s_1 s_2 s_3^4 + 2s_{15} a s_1 s_2 x_2^2 s_9 + s_{15} b s_{16}^5 + 2s_{15} a s_1 s_2 s_{11} + s_{15}^3 b x_1 + s_{15} b x_1 s_4 \tag{38}$$